ECHO CHAMBER BREAKER

Complete UI/UX & Interface System Specification

Primary Experience • Navigation • Hidden Customization Studio • AWS + Alexa+ Standards

Document Type	UI/UX System Spec (normative for implementation)
Primary Goal	Clean, focused intellectual-sparring experience
Customization Policy	All advanced controls hidden from main view
Audience	Frontend developers, Antigravity agents, UX reviewers
Alignment	AWS Bedrock + Alexa+ + MCP Streamable HTTP

This specification defines the complete interface system. The main view is deliberately minimal and focused on the intellectual honesty loop. All customization, theme, voice, profile, and advanced cognitive controls live in a secondary Customization Studio that is not visible during normal use.

1. Design Principles

P1	**Focus over features**	The main screen exists for one job: express a conviction → receive a steelman → engage. Everything else is secondary.
P2	**Customization is hidden**	Themes, rigor intensity, cognitive lens, voice settings, and profiles never appear on the primary sparring surface.
P3	**Authentic protocol signal**	Live MCP status and protocol trace remain visible (or one click away) because they prove the system is real for judges and developers.
P4	**Voice-first secondary surfaces**	Any content that Alexa may speak must have a short spoken form. Long text is for screens.
P5	**Explainable score**	Honesty score changes are always accompanied by a visible ledger reason. No silent mutations.
P6	**Dark, calm, serious**	The product is an intellectual sparring partner, not a playful consumer app. Visual language is restrained.
P7	**No invented conviction in UI**	The interface must never show a Cognitive Stance Diagnosis for non-stance input (see Logic System).

2. Information Architecture & Navigation

2.1 Top-Level Surfaces

Surface	Purpose	Access
Main Sparring View	Primary experience: input → diagnosis → steelman → engagement → score	Default / home
Honesty Ledger Panel	Score, radar, audit trail, Challenge Me	Always visible on right (desktop) or tab (mobile)
Live MCP Trace	Real protocol traffic for authenticity	Collapsible panel; open by default in demo mode
Customization Studio	All advanced controls (theme, rigor, lens, voice, profiles)	Hidden — opened only via explicit entry point
Export / Certificate	Shareable honesty summary or demo artifact	Header action

2.2 Header (Global)

Persistent top bar. Minimal. Never overloaded with customization controls.

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[ Logo + Product Name ]  [ MCP Status Badge ]  [ Voice Status ]  [ Customizations ■ ]  [ Export ]
Subtitle: short product line (Anti-Sycophant Dialectical Partner)
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The only permanent entry to advanced settings is the single “Customizations” control (gear or labeled button). It opens the Customization Studio as a full overlay or secondary route — never inline on the main view.

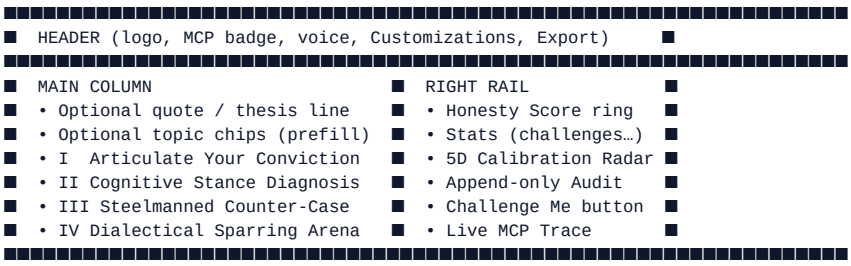
2.3 Navigation Model

- No traditional multi-page app navigation for the core loop. The main view is a single progressive disclosure surface.
- Sections on the main view are numbered (I Articulation → II Diagnosis → III Steelman → IV Arena) and reveal as the user progresses.
- Customization Studio is a distinct mode. User must explicitly leave it to return to sparring.
- Topic example chips may appear above the input as optional pre-fills only. They do not constitute navigation.

3. Main Sparring View (Primary Experience)

This is the only screen most users and judges should spend time on. It must feel calm, serious, and purposeful.

3.1 Layout Zones (Desktop)



3.2 Section-by-Section Specification

I — Articulate Your Conviction

- Large text area for the user's statement. Placeholder invites a real opinion.
- Primary actions: Analyze Stance | Steelman Opposing View | Multi-Turn Debate.
- Secondary: Speak to Alexa (if voice enabled).
- Tool call indicator (e.g. tools/call: analyze_stance) may appear for developer/demo credibility.
- Never show a diagnosis until Stance Gate has accepted the input as a real stance.

II — Cognitive Stance Diagnosis

- Only visible after successful analyze_stance with is_stance=true.
- Shows: Polarity, Conviction Certainty bar, Emotional Charge bar.
- Lists Identified Cognitive Blind Spots (specific, non-generic).
- Shows one Unexamined Assumption sentence.
- Must clear completely if a subsequent non-stance input is submitted.

III — The Steelmanned Counter-Case

- Alexa Spoken Synthesis block (≤40 words) highlighted.
- Foundational Counter-Thesis + Core Structural Premises.
- Socratic Inquiry question.
- Engagement action row: Dismiss (–pts) | Concede Valid Point (+pts) | Submit Counter-Rebuttal (+pts).
- These actions are the only triggers for score mutation from this surface.

IV — Dialectical Sparring Arena

- Appears when Multi-Turn Debate is started.
- Turn history (You / EchoBreaker) with clear visual distinction.
- Input for next user turn + Submit Turn / Conclude Debate.
- Session status badge (Active / Concluded).

4. Right Rail — Honesty Ledger & Protocol

4.1 Intellectual Honesty Score

- Large ring or numeric display of current score.
- Derived label under the number (e.g. CONFIRMATION_TRAPPED, CALIBRATING, OPEN_MINDED) based on score band + recent pattern.
- Never the only source of truth — score is always derived from the append-only ledger.

4.2 Stats Strip

Challenges · Engagement % · Concessions (or equivalent high-signal counters).

4.3 5D Epistemic Calibration Radar

- Axes example: Openness, Rigor, Humility, Rebuttal Quality, Consistency.
- Values must be driven by real engagement data, not decorative.
- Tooltip or legend available on hover/focus.

4.4 Append-Only Audit Trail

- Newest events at top.
- Each row: short reason + delta + resulting score.
- Tamper-proof / append-only visual treatment.
- Clicking an event may expand details (optional).

4.5 Challenge Me

Proactive entry point. Calls challenge_me tool. Should respect fatigue (do not spam the user with challenges).

4.6 Live MCP Protocol Trace

- Collapsible. Recommended open in demo / judge mode.
- Shows real JSON-RPC request/response traffic.
- Proves the simulator is a real MCP client.
- Clear visual distinction between client → server and server → client messages.

5. Customization Studio (Hidden from Main View)

All advanced controls live here. This surface is never the default. It is opened only when the user explicitly chooses Customizations from the header.

5.1 Access Rule (Normative)

No theme switcher, rigor slider, cognitive lens, voice pitch control, or profile selector may appear on the Main Sparring View. Violation of this rule breaks the Focus principle (P1/P2).

5.2 Studio Sections

Section	Controls	Notes
Visual Theme & Palette	Amazon Cloud / Alexa Echo / AWS Console / Classic Socratic	Affects colors only; never layout logic
Sparring Rigor Intensity	Gentle Inquiry / Standard Socratic / Ruthless Stress-Test	Maps to system prompt intensity for steelman
Cognitive Perspective Lens	Empirical Economist / First-Principles / Contrarian Maverick / Devil's Advocate	Influences framing of opposing case
Alexa Voice & Sound	Speed, Pitch, Echo Chimes on/off	Only relevant when voice synthesis is used
Device & Profile Dossier	Default Living Room / Enterprise CTO / Public Policy Analyst + Switch Profile	Optional persona context for challenge generation

5.3 Studio UX Rules

- Studio opens as a full-width panel or dedicated route, not a tiny popover.
- Clear "Back to Sparring" or close control that returns the user to the exact previous main-view state.
- Changes to rigor / lens take effect on the next steelman or challenge, not retroactively on already rendered content.
- Theme changes may apply immediately for visual feedback.
- Studio must not itself run analyze_stance or steelman. It only configures future behavior.

6. Interface States & Empty / Error Handling

State	UI Behavior
Fresh / Idle	Input empty or placeholder. Diagnosis, Steelman, Arena sections collapsed or hidden. Score shows baseline.
Non-stance reply	Soft message in place of diagnosis. No polarity bars. No steelman offer.
Analyzing	Subtle loading on Analyze action. Prevent double-submit.
Diagnosis ready	Section II expands with data. Steelman button becomes primary.
Challenge active	Section III visible with engagement actions. Score rail updates only after action.
Debating	Section IV expands. Session Active badge. Turn list grows.
MCP offline	Header badge turns error. Trace shows disconnect. Actions that need tools are disabled with clear reason.
Tool error / timeout	Inline error near the action. Never leave a previous unrelated steelman visible as if it were the new result.

7. Component & Interaction Rules

7.1 Buttons & Actions

- Primary action (Steelman when ready) uses the accent/gold treatment.
- Destructive or cost actions (Dismiss) use a restrained danger treatment, not aggressive red if possible.
- Positive engagement (Concede) uses a success/teal treatment.
- Disabled state must explain why (tooltip or helper text) when tools are unavailable.

7.2 Progressive Disclosure

- Do not show empty Diagnosis or Steelman shells before data exists.
- Reveal sections only when the preceding step has valid data.
- Avoid large blank cards that make the product look unfinished.

7.3 Topic Chips

- Optional. Purpose: one-click example statements for demo and exploration.
- Selecting a chip only pre-fills the text area. It does not run analysis by itself unless the user then clicks Analyze.
- Analysis always uses the current text in the box, never the chip label alone.

7.4 Score Feedback

- When a ledger event is written, the score ring and audit trail update immediately.
- A short transient highlight on the new audit row is recommended.
- Do not animate the score in a gamified or celebratory way for Dismiss (cost) actions.

8. Responsive Behavior & Voice Considerations

8.1 Breakpoints (Recommended)

- Desktop (≥1100px): two-column (main + right rail).
- Tablet: right rail becomes a bottom sheet or secondary tab (Ledger | Trace).
- Mobile: single column; Ledger and Trace accessible via bottom navigation or header icons.

8.2 Voice / Alexa Alignment

- Every tool that may be spoken exposes a say field. UI should surface the spoken version clearly when relevant.
- “Speak to Alexa” / “Alexa Audio” controls indicate voice path availability.
- Long steelman content is screen-oriented; the spoken synthesis is the short form.
- Customization of speed/pitch lives only in the Studio.

9. Developer & AWS/Alexa+ Alignment

The interface must continuously signal that this is a real self-hosted MCP server powered by Amazon Bedrock, not a mocked frontend demo.

9.1 Trust Signals (Always Appropriate on Main View)

- MCP Streamable HTTP status badge (Active / Offline).
- Live MCP Protocol Trace (collapsible).
- Optional small “AWS Bedrock” and “Alexa+ Cognitive Layer” tags in the header subtitle area.
- Tool-call name indicators near actions (tools/call: analyze_stance) during demo.

9.2 What Belongs Only in Studio or Docs

- Model ID selection, temperature, or raw Bedrock parameters.
- Full system prompt editing.
- Theme and persona experimentation.
- Export certificate generation details.

9.3 Demo Mode Recommendation

For hackathon judging, provide a Demo Mode toggle (can live in Studio or a subtle header control) that: keeps the MCP Trace open, uses a polished example stance on first load if desired, and ensures the score ledger is visible. Demo Mode must still call the real MCP server.

10. UI/UX Acceptance Checklist

- ■ Main Sparring View contains no theme, rigor, lens, voice-pitch, or profile controls.
- ■ Customizations are reachable only via a single explicit header entry and open a separate Studio surface.
- ■ Non-stance input never reveals a Cognitive Stance Diagnosis panel.
- ■ Steelman and engagement actions only appear after a valid stance diagnosis.
- ■ Score changes only after Dismiss / Concede / Rebuttal / Debate conclusion (or equivalent).
- ■ Live MCP Trace shows real traffic; UI never invents tool results.
- ■ Topic chips only pre-fill text; analysis always uses the current text area content.
- ■ Progressive disclosure: empty diagnosis/steelman shells are not shown before data exists.
- ■ Header remains minimal: logo, status badges, Customizations entry, Export.
- ■ Studio has a clear path back to Sparring without losing the user’s place.
- ■ Voice “say” content is visible where Alexa synthesis is relevant.
- ■ Error and offline states are explicit and do not leave stale steelman content on screen.

The interface succeeds when a new user can articulate a conviction, receive a serious opposing case, and engage — without ever needing to open Customizations. The Studio exists for power users and demo polish, not for the core loop. Keep the main view clean. Hide the complexity.

Echo Chamber Breaker — Interface: focus first, customization second.